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Whole cottonseed and palmitic and oleic acid supplementation improve production responses during the immediate postpartum in multiparous dairy cows

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ABSTRACT

We evaluated the effects of whole cottonseed (WCS) and fatty acid (FA) supplementation on the production responses of early-lactation cows. Fifty-two multiparous cows were used in a randomized complete block design with a 2×2 factorial arrangement of treatments. The following treatment diets were fed from 1 to 21 d postpartum (fresh period; FR): (1) a diet containing no WCS or supplemental FA (CON); (2) a diet containing WCS at 10% diet DM (CT); (3) a diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9 (FAT); and (4) a diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt (CT+FAT). In the WCS-containing diets, WCS replaced soybean hulls and soybean meal. The FA supplement was added at 1.9% of diet DM to deliver ~1.5% of FA and replaced soyhulls in the CT and CT+FAT treatments. During the carryover period (CO; 22–42 d postpartum), all cows were offered a common diet to evaluate the carryover effects of the treatment diets. During the FR period, WCS and FA supplementation interacted with time to affect plasma nonesterified fatty acids. During FR period, WCS and FA supplementation interacted to affect DMI, NDF intake, the yields of milk fat and 3.5% FCM, and plasma BHB, and tended to interact to modify the yield of ECM and milk energy output. The CT treatment increased DMI compared with CON and FAT. The CT, FAT, and CT+FAT treatments increased the yields of 3.5% FCM and milk fat and the cumulative milk fat and tended to increase the yield of ECM and milk energy output compared with CON. However, when WCS and the supplemental FA were fed together, no further improvement was observed in milk production responses compared with when they were fed separately. Whole cottonseed and time tended to interact to modify plasma insulin. Overall, WCS increased DMI, most of the milk production responses, en-

ergy intake, and plasma gossypol; improved energy balance; reduced 18-carbon and total FA digestibility; and tended to reduce NDF digestibility. Overall, FA increased feed efficiency, milk fat content, and plasma gossypol; tended to increase milk fat yield and cumulative milk fat yield; and tended to reduce milk protein content. We did not observe treatment differences for BW, BCS, BW change, and BCS change during the FR period. During the CO period, WCS and FA tended to increase cumulative protein yield, and FA tended to reduce BW loss. The absence of a synergistic or additive effect when feeding both WCS and supplemental FA could be attributed to their lack of effect on DMI and reduced digestibility of NDF and FA. In conclusion, these results suggest that both WCS and supplemental FA in the diet can enhance milk production in early-lactation dairy cows, with WCS improving milk fat and protein yields and supplemental FA improving milk fat yield.

Key words: supplemental, whole cottonseed, fatty acid, early lactation

INTRODUCTION

During the immediate postpartum period, high-producing dairy cows experience a period of negative energy balance because DMI is insufficient to meet the increasing nutrient requirements for lactation (van Knegsel et al., 2007). Consequently, cows mobilize body energy reserves, releasing nonesterified fatty acids (NEFA) into circulation to be metabolized for energy by the liver and other tissues and incorporated into milk fat in the mammary gland (Drackley, 1999). However, intense body energy reserve mobilization and persistent negative energy balance increase the risk of metabolic disorders and poor animal performance (Allen and Piantoni, 2013). Some approaches used to increase energy intake postpartum include supplemental fatty acids (FA) and the inclusion of oilseeds in diets (Rabiee et al., 2012; Lock et al., 2025).

Previous recommendations advised against increasing dietary FA in early-lactation cows (Grummer, 1992; Hutjens, 1996) due to possible negative effects on DMI

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and the endocrine system (Kuhla et al., 2016). However, recent studies have reported that FA supplementation can have a positive effect on production responses. In early-lactation cows, supplemental palmitic acid (C16:0) increased the yield of ECM, tended to increase BW loss, and did not modify DMI compared with a control diet (de Souza and Lock, 2019), while supplementing C16:0 alone or with chromium increased milk production responses without affecting BW loss (Parales-Girón et al., 2025). Feeding FA blends with different ratios of C16:0 and oleic acid (C18:1 *cis*-9) during the first 3 wk of lactation increased DMI and the yields of milk, milk fat, and ECM compared with a nonsupplemental fat control diet, with increasing content of C18:1 *cis*-9 linearly reducing BW loss (de Souza et al., 2021; Abou-Rjeileh et al., 2025). These results demonstrate that FA supplementation can positively affect dairy cow metabolism, and specific FA may alter energy partitioning differently.

Some recent research in early-lactation cows has focused on FA supplements to increase the energy density of the diet. Oilseeds, however, represent an alternative source of FA that can increase dietary FA content and have often been reported to improve milk production in established lactation (Rabiee et al., 2012; Bales and Lock, 2024; Bales et al., 2024a). Whole cottonseed (WCS) is a common byproduct used in dairy cows' diets because of its high content of FA and CP (Moreira et al., 2004) and its physically effective fiber. Feeding WCS increased the yields of milk fat and 3.5% FCM compared with control diets with no WCS in mid-lactation dairy cows (Rico et al., 2017; de Souza et al., 2018). Furthermore, increasing dietary content of WCS up to ~15% DM improved milk production responses of high-producing cows (Bales et al., 2024a; Adeniji et al., 2025). Responses to feeding WCS in early lactation (after 21 d of lactation) have been inconsistent, including no significant effect on milk production responses (Chen et al., 2008). However, some studies have shown positive effects on milk yield without any effects on milk component yields (Abel-Caines et al., 1997). Additionally, multiparous cows fed WCS had increased milk fat and ECM yields (Harrison et al., 1995), but no further improvements were observed when WCS was fed alongside calcium salt of long-chain FA at a concentration of 2.7% of DM (Harrison et al., 1995). Whereas the available literature documents the effect of feeding WCS after 21 d postpartum, there are no reports, to our knowledge, on the effects of feeding WCS during the immediate postpartum period on production responses.

We have previously examined the interactions between FA supplements and low and high basal diets with or without WCS, obtaining variable results (Rico et al., 2017; de Souza et al., 2018; Bales et al., 2024b). However, as far as we know, no studies in the immediate postpartum have

evaluated the interaction between WCS and FA supplementation. Therefore, the objective of our study was to evaluate the effects of WCS feeding and FA supplementation on nutrient digestibility, milk production responses, and energy balance of early-lactation dairy cows. We hypothesized that supplemental FA would increase energy partitioning toward milk without increasing body energy reserves mobilization, WCS would increase milk and milk fat yields, and supplemental FA plus WCS would yield further improvement in milk production responses. To keep the nutrient composition of the diets similar, WCS replaced soybean meal and soyhulls in WCS-supplemented diets. Thus, although treatment diets contained similar contents of NDF, forage NDF, CP, and starch, they differed in FA content.

MATERIALS AND METHODS

Animal Housing and Care

The Institutional Animal Care and Use Committee at Michigan State University (East Lansing, MI) approved all experimental procedures. The experiment began on September 11, 2022, and ended on May 20, 2023. Cows were fed once daily (0800 h) at 115% of expected intake during the fresh (FR) and carryover (CO) periods, and they were milked 3 times daily (0400, 1200, and 2000 h). Water was available ad libitum in each stall, and stalls were bedded with sawdust and cleaned twice daily. Standard reproduction and health herd checks, as well as breeding practices, were maintained throughout the study.

Design and Treatment Diets

Fifty-two multiparous Holstein cows at the Michigan State University Dairy Cattle Teaching and Research Center (East Lansing, MI) were used in a randomized complete block design experiment with a 2×2 factorial arrangement of treatments. Cows were blocked by previous lactation 305-d mature-equivalent milk production (mean $\pm$ SD; $13,535 \pm 1,435$ kg), BCS (up to 0.50-unit difference using a scale of 1 = thin and 5 = fat in increments of 0.25, BCS 3.36 ± 0.19), and expected parturition date (40 ± 25 d). The BCS used to block cows was the measurement recorded before parturition. Cows within each block were randomly assigned to treatment on their enrollment day, 21 d before their expected parturition date. The number of cows per treatment ($n = 13$) was chosen based on previous early-lactation studies evaluating 2 levels of NDF and FA in a 2×2 factorial design (Piantoni et al., 2015) and 2 levels of chromium and FA supplementation in a 2×2 factorial design (Parales-Girón et al., 2025).

Treatments were (1) a diet containing no WCS or supplemental FA (CON); (2) a diet containing WCS (CT); (3) a diet supplemented with a calcium salt containing (% of total FA) 60% C16:0 and 30% C18:1 *cis*-9 (FAT); and (4) a diet containing WCS and the C16:0 C18:1 *cis*-9 calcium salt (CT+FAT). The inclusion level of the FA supplement was based on previous studies evaluating increasing levels of C16:0 in the diets of mid-lactation cows (Rico et al., 2017), as well as studies assessing the inclusion of C16:0 and C16:0 and C18:1 *cis*-9 supplements in early-lactation cows (de Souza and Lock, 2019; de Souza et al., 2021). The inclusion level of WCS was selected to provide a comparable amount of supplemental FA to that supplied by the FA supplement. The FA supplement (calcium-salts) used in the FAT and CT+FAT treatments was formulated as a 60:30 blend of C16:0 and C18:1 *cis*-9, but our analysis showed actual concentrations of 67.3% and 21.2%, respectively (Table 1). In the WCS-containing diets, WCS was added at 10% of diet DM and replaced soyhulls and soybean meal to keep NDF and CP levels similar across all diets. The FA supplement (Spectrum Distinct, Perdue Agribusiness, Salisbury, MD) was added at 1.9% of diet DM to deliver ~1.5% of FA and replaced soyhulls in the FAT and CT+FAT treatments. Treatment diets were offered from 1 to 21 d postpartum during the FR period, and all cows were fed a common diet to evaluate possible carryover effects of the treatment diets from 22 to 42 d postpartum (CO period). Treatment diets were mixed daily in a tumble-mixer and fed from the morning following parturition, and common diets (close-up and carryover) were mixed daily in a mixer wagon. The treatment and carryover diets were adjusted for NDF, CP, and starch concentrations. The DM concentration of forages was determined twice weekly using a Koster Moisture Test (Koster Moisture Testers Inc.), and diets were adjusted when necessary. The ingredient and nutrient composition of the diets fed as TMR, including the close-up ration for reference, are described in Table 1. All rations were formulated using CNCPS v. 6.5.5 (AMTS.Cattle.Professional v. 4.17.0.0; AMTS LLC; Van Amburgh et al., 2015). Feed access was blocked from 0600 to 0800 h for orts collection and offering of new feed. The acronyms CON, CT, FAT, and CT+FAT are used only in reference to the control, WCS treatment, FA treatment, and WCS + FA treatment, respectively.

Data and Sample Collection

Milk yield and feed offered and refused were recorded daily throughout the experiment. All samples and body measurements were collected or recorded on the same days of the week. Therefore, there was a variation of ± 1 d relative to the average DIM reported below for data

and sample collection. Samples of all diet ingredients (0.5 kg) and orts from each cow (~12.5%) were collected weekly during the FR period, and diet ingredients were collected monthly during the CO period and stored in plastic bags at -20°C until being processed. Milk samples were collected twice a week at each milking during the FR period and once a week during the CO period and stored with preservative (Bronolab W-II liquid, Advanced Instruments, Norwood, MA) at 4°C for component analysis. During the FR period, collection days for milk components were d 4 and 7, 11 and 14, and 18 and 21 postpartum, with the averages from these sets of days representing wk 1, 2, and 3, respectively. During the CO period, collection days for milk components were d 28, 35, and 42, representing wk 4, 5, and 6, respectively. An additional milk sample was collected at each milking on d 7, 14, and 21 postpartum and stored without preservative at -20°C for FA analysis.

On d 21 postpartum, a fecal sample (0.5 kg) was collected every 6 h for a 24-h period to account for diurnal variation and used for nutrient digestibility analysis. Feces were stored in sealed plastic cups at -20°C until dried, ground, and composited per cow per sampling day. Blood samples were collected weekly during the FR period on d 7, 14, and 21 postpartum by puncture of coccygeal vessels 1 h before feeding. Blood was collected into 2 evacuated tubes containing potassium EDTA. Both tubes were centrifuged at $3,000 \times g$ for 15 min immediately after sample collection, and plasma was harvested and stored at -20°C until analysis. One additional plasma aliquot on d 21 and 40 was immediately stored at -80°C for gossypol analysis. Body weight was recorded weekly from d -21 of the expected parturition day until calving and 3 times per week after calving. Body condition score was recorded once 3 wk before the expected calving date and weekly after calving by 3 trained investigators using a 5-point scale, as described by Wildman et al. (1982). The BW (kg/d) and BCS changes (units/wk) were calculated for each cow within the period by linear regression as described by Boerman et al. (2015).

Apparent NEL concentration of the diet was calculated from digestibility results as described by Harvatine and Allen (2006). Energy intake was calculated for individual cows for each treatment from apparent NEL of diet $\times$ DMI (NRC, 2001). Energy for maintenance, milk energy output, and mobilization of body energy reserves was calculated according to NASEM (2021). Energy balance (Mcal/d) was determined as NEL intake (Mcal/d) $-$ milk NEL (Mcal/d) $-$ NEL maintenance requirement (Mcal/d; NASEM, 2021).

Table 1. Ingredient and nutrient composition of close-up, treatment, and carryover diets

Item	Diet					
	Close-up	Treatment ¹				Carryover
		CON	CT	FAT	CT+FAT	
Ingredient, % DM						
Corn silage	46.6	33.0	33.1	32.9	33.0	40.4
Alfalfa silage	—	12.1	12.0	12.2	11.8	13.9
Alfalfa hay	—	6.49	6.36	6.25	6.49	—
Grass hay	15.7	—	—	—	—	—
Straw	13.3	—	—	—	—	—
Ground corn	0.87	15.1	14.9	14.9	15.1	20.8
Soybean meal	4.58	11.4	8.49	11.4	8.50	10.7
Soyhulls	4.33	12.7	5.83	11.1	4.10	5.49
Whole cottonseed	—	—	10.1	—	9.93	—
DCAD supplement close-up ²	7.38	—	—	—	—	—
DCAD supplement ³	—	0.32	0.30	0.35	0.29	0.51
Protein supplement ⁴	—	6.78	6.84	6.77	6.82	6.28
FA supplement ⁵	—	—	—	1.90	1.85	—
Mineral, vitamin, and protein mix close-up ⁶	7.26	—	—	—	—	—
Mineral and vitamin mix ⁷	—	2.16	2.17	2.11	2.17	2.04
Nutrient composition, ⁸ % DM						
NDF	42.6	31.3	30.7	30.0	29.6	27.1
Forage NDF	36.2	20.2	19.9	20.1	20.0	20.2
CP	13.2	17.6	17.6	17.4	17.3	16.6
Starch	17.3	23.6	23.5	23.4	23.5	30.0
FA	1.25	1.67	3.06	3.17	4.51	1.93
C16:0	0.27	0.27	0.60	1.29	1.59	0.30
C18:0	0.04	0.05	0.08	0.12	0.14	0.05
C18:1 <i>cis</i> -9	0.24	0.30	0.52	0.62	0.84	0.37
C18:2 <i>cis</i> -9, <i>cis</i> -12	0.59	0.85	1.64	0.91	1.70	1.01
C18:3 <i>cis</i> -9, <i>cis</i> -12, <i>cis</i> -15	0.08	0.15	0.14	0.15	0.14	0.16

¹CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

²Soychlor (West Central Soy, Ralston, IA).

³Electrolyte mix contained (DM basis) 48.5% potassium and 0.31% sodium (Ion Plus; D&D Ingredients LLC, Delphos, OH).

⁴Protein supplement contained 40% sodium sesquinate, 20% calcium carbonate, 19% ground corn, 15% blood blend (Spectrum AgriBlue; Perdue Agribusiness, Salisbury, MD), 3% rumen-protected Met (Smartamine M; Adisseo, Alpharetta, GA), and 2% urea.

⁵Calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9 (Spectrum Distinct; Perdue Agribusiness, Salisbury, MD). The supplement contained (g/100 g FA) 1.56 of C14:0, 67.3 of C16:0, 4.52 of C18:0, and 21.2 of C18:1 *cis*-9, and 79.7% total FA.

⁶Mineral, vitamin, and protein mix used in close-up diet contained (DM basis) 77.1% heat-treated soybean meal (Amino Plus; Ag Processing Inc., Mason City, IA), 8.47% magnesium sulfate salt, 7.80% blood blend (Spectrum Agriblue; Perdue Agribusiness, Salisbury, MD), 2.56% animal fat, 1.34% magnesium oxide, 1.88% white salt, and 0.85% trace minerals and vitamins.

⁷Mineral and vitamin mix contained 26% bypass protein (Amino Plus, Ag Processing Inc. Mason City, IA); 14% calcium carbonate; 13% blood meal (Caledonia Pass, Papillon) and sodium sesquicarbonate; 9% calcium phosphate; 6% urea; 5% white salt; 4% magnesium oxide; 3% calcium magnesium carbonate (MIN AD; Min Ad Inc., Winnemucca, NV) and potassium chloride red; 2% potassium electrolyte mix (Ion Plus; D&D Ingredients LLC, Delphos, OH); 1% selenium; and <1% ferrous sulfate, trace minerals, (Smartamine M; Adisseo, Alpharetta, GA), vitamin D, vitamin E, and vitamin A.

⁸Analysis of CP, NDF, and starch performed at Cumberland Valley Analytical Services, Inc. (Waynesboro, PA) as described by Boerman et al. (2017), and FA analysis performed internally as described by Bales and Lock (2024).

Sample Analysis

Feed, orts, and feces samples were dried in a 55°C forced-air oven for 72 h, and DM content was determined. Before drying, feed ingredients were composited by month. Orts from individual cows were dried to calculate

weekly DMI during the FR period. Once dried, samples of feed ingredients, feces, and orts were ground with a Wiley mill (1-mm screen; Arthur H. Thomas Co., Philadelphia, PA) and analyzed for NDF, indigestible NDF, starch, and CP according to Boerman et al. (2017) and FA according to Bales et al. (2024c). Indigestible NDF

(also referred to as uNDF240 in some literature) was used as an internal marker to estimate fecal output, which was used to calculate apparent total-tract digestibility of nutrients (Cochran et al., 1986). Indigestible NDF was estimated as the fraction of NDF that remains undigested after 240 h of in vitro fermentation (Goering and Van Soest, 1970). To maintain a viable microbial population, the flasks were reinoculated at 120 h of fermentation. Nutrient total-tract digestibility was calculated using total-tract disappearance.

Milk samples were analyzed for fat, true protein, and lactose concentrations by mid-infrared spectroscopy (Association of Official Analytical Chemists, 1990; method 972.160) by the Michigan Dairy Herd Improvement Association (CentralStar Michigan Lab, Grand Ledge, MI). Yields of 3.5% FCM, ECM, and milk components were calculated using milk yield and component concentrations from each milking, summed for a daily total, and averaged for each week. Milk lipids were extracted, and FA-methyl esters were prepared and quantified using GC, according to Bales and Lock (2024). Yields of individual FA (g/d) in milk fat were calculated using milk fat yield and FA concentrations to determine yield on a mass basis using the molecular weight of each FA while correcting for glycerol content and other milk lipid classes (Benoit et al., 2024). Plasma samples were analyzed at the Michigan State University Veterinary Diagnostic Laboratory (Lansing, MI) for concentrations of NEFA and BHB using an Olympus AU640e chemistry analyzer (Olympus America, Center Valley, PA), and plasma insulin was quantified with a bovine insulin ELISA using a solid-phase 2-site enzyme immunoassay (Bovine Insulin ELISA; Mercodia AB, Uppsala, Sweden). Plasma gossypol was analyzed for total and (–) and (+) isomers and WCS for total gossypol using the method described by Bullock et al. (2010) based on AOCS Recommended Practice BA 8a-99 (AOCS, 1998) at the USDA-ARS, Southern Regional Research Center (New Orleans, LA).

Statistical Analysis

Data were analyzed separately for the FR and CO periods. All weekly data were analyzed using the GLIMMIX procedure of SAS v.9.4 (SAS Institute, Inc., Cary, NC), with week of lactation being the repeated measures. The model used included:

$$Y_{ijklmn} = \mu + F_i + L_k + T_1 + F_iL_k + F_iT_1 + L_kT_1 + F_iL_kT_1 + B_m + C(B_mF_iL_k)_j + J_n + e_{ijklmn},$$

where μ = overall mean; F_i = fixed effect of WCS; L_k = fixed effect of FA supplementation; T_1 = fixed effect of time; F_iL_k = fixed effect of the interaction between WCS

and FA; F_iT_1 = fixed effect of the interaction between WCS and time; L_kT_1 = fixed effect of the interaction between FA and time; $F_iL_kT_1$ = fixed effect of the interaction between WCS, FA, and time; B_m = random effect of block; $C(B_mF_iL_k)_j$ = random effect of cow within block and treatment diet; J_n = random effect of Julian date; and e_{ijklmn} = residual error. A reduced model without the effects of time and the effect of cow within block and treatment was used to determine the effect of treatments on variables that were recorded once per period. First-order autoregressive was the covariate structure used for analysis because it resulted in the smallest Bayesian information criterion for most measured variables. Precalving BCS was used as a covariate for BCS and BCS change. Plasma gossypol data lacked a normal distribution and were transformed using a $\log_{10}$ function, and back-transformed values were reported. Main effects were declared significant at $P \leq 0.05$ and tendencies at $P \leq 0.10$, and interactions were declared significant at $P \leq 0.10$ and tendencies at $P \leq 0.15$. When 3-way or 2-way interactions were $P \leq 0.15$, the SLICE option in the LSMEANS statement was used to evaluate treatment effects within a week of lactation or the effect of FA treatments within WCS treatments, respectively. Although no significant effects or tendencies were observed ($P \geq 0.15$), we sliced and plotted DMI, milk yield (Figure 1), and the yields of milk protein and milk lactose (Figure 2) to illustrate the behavior of these variables over time across the whole experiment. All cows were in apparent good health at the beginning of the study, and treatment groups were not different in terms of 305-d mature-equivalent yield ($13,535 \pm 1,465$ kg; $P = 0.77$), BW (758 ± 64 kg; $P = 0.66$), or BCS (3.35 ± 0.19 ; $P = 0.96$) precalving. One of the cows on the CT+FAT treatment had digestive issues for most of the CO period. Therefore, her data for the CO period were excluded from the statistical analyses.

RESULTS

Health Events

A summary of health events is presented in Supplemental Table S1, see Notes. During the FR period, 10 health events were recorded; retained placenta was the primary health event observed, followed by ketosis. The number of health events was lower during the CO period, with 3 events.

Nutrient Intake and Digestibility During the FR Period

Whole cottonseed and FA supplementation interacted to alter DMI ($P = 0.08$; Figure 1A), NDF intake ($P = 0.09$), 16-carbon FA intake ($P < 0.01$), and 16-carbon

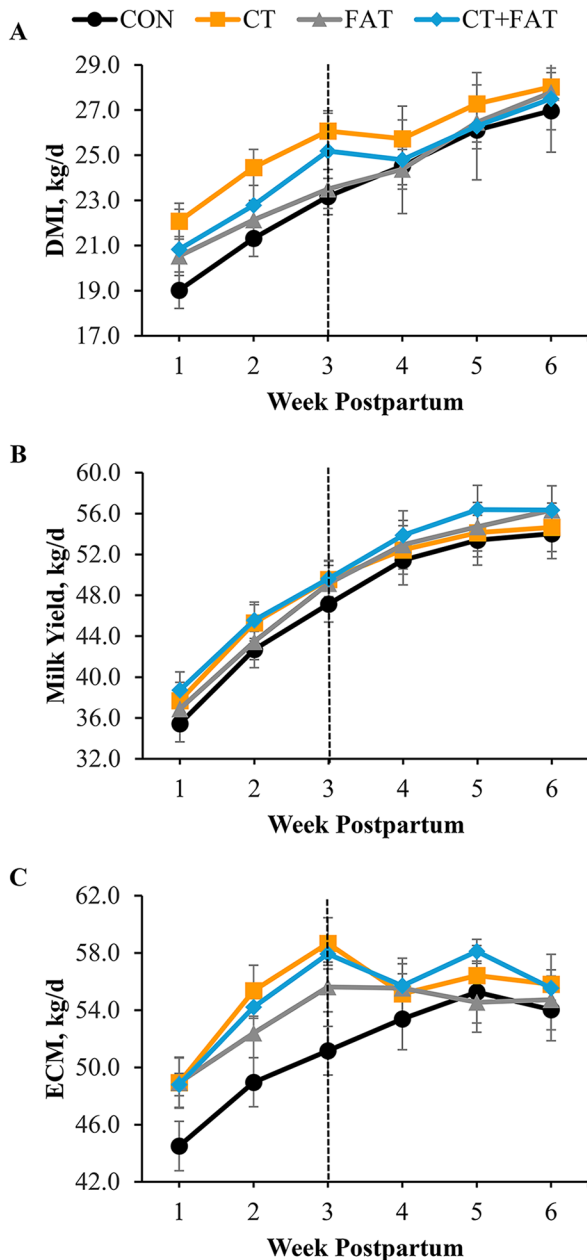


Figure 1. Effects of dietary treatments on DMI (A), milk yield (B), and ECM yield (C) during the FR (d 1–21 postpartum) and CO (d 22–42 postpartum) periods. Diets fed during the FR period were CON (a diet containing no WCS or supplemental FA; black line); CT (diet containing WCS at 10% diet DM; orange line); FAT (diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; gray line); and CT+FAT (diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt; blue line). The vertical dashed line on wk 3 indicates the start of the CO period, when all cows were fed a common diet with no WCS or supplemental FA. During the FR period, WCS and FA interacted to affect DMI ($P = 0.08$). No treatment effects were observed for DMI during the CO period ($P \geq 0.38$). No treatment effects were observed for milk yield during the FR and CO periods ($P \geq 0.19$ and $P \geq 0.22$). The WCS and FA interacted to affect ECM yield during the FR period ($P = 0.13$). During the CO period, WCS and FA interacted with time to affect ECM yield ($P = 0.03$), but when treatments were compared within week of lactation, no differences were observed ($P > 0.10$). Error bars indicate SEM.

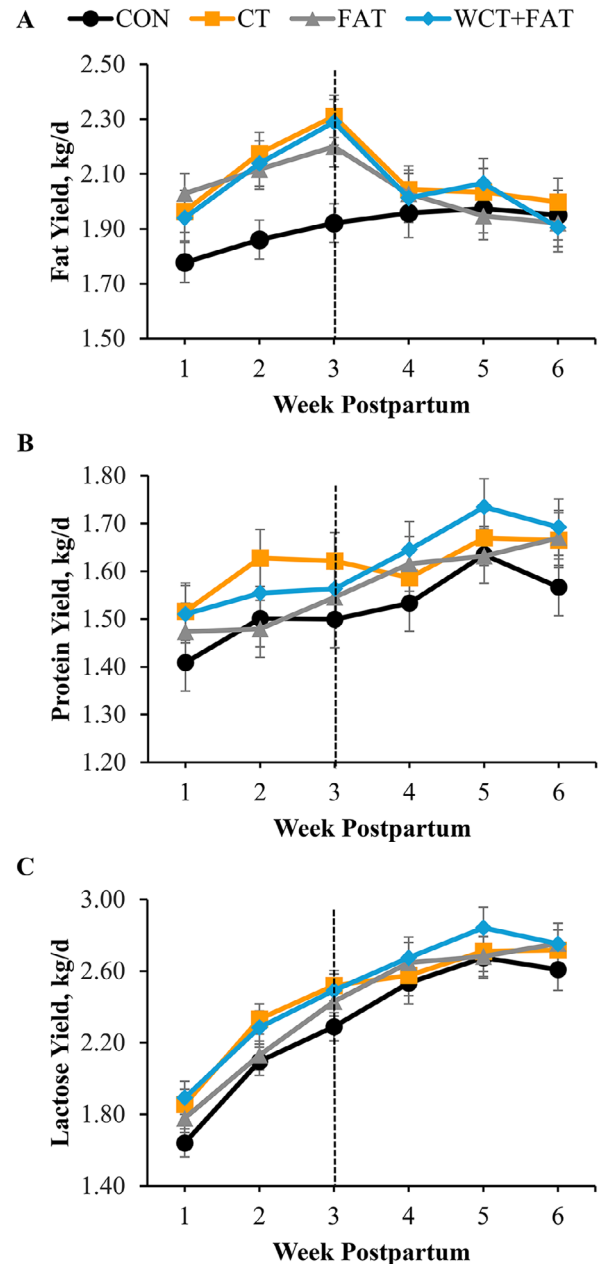


Figure 2. Effects of dietary treatments on milk fat yield (A), milk protein yield (B), and milk lactose yield (C) during the FR (d 1–21 postpartum) and CO (d 22–42 postpartum) periods. Diets fed during the FR period were CON (a diet containing no WCS or supplemental FA; black line); CT (diet containing WCS at 10% diet DM; orange line); FAT (diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; gray line); and CT+FAT (diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt; blue line). The vertical dashed line on wk 3 indicates the start of the CO period, when all cows were fed a common diet with no WCS or supplemental FA. During the FR period, WCS and FA interacted to influence the milk fat yield ($P = 0.03$). No treatment effects were observed for milk fat yield during the CO period ($P \geq 0.19$). During the FR period, WCS increased milk lactose yield ($P = 0.03$) and tended to increase milk protein yield ($P = 0.10$). During the CO period, WCS, FA, and time interacted to influence the yields of milk protein ($P = 0.01$) and milk lactose ($P < 0.01$), but when treatments were compared within week of lactation, no differences were observed ($P \geq 0.10$). Error bars indicate SEM.

FA absorption ($P = 0.01$; Table 2). The CT treatment increased DMI compared with CON and FAT and increased NDF intake compared with the other 3 treatments (Table 2). Relative to CON, CT, and FAT, CT+FAT increased 16-carbon FA intake (Table 2). We did not observe any interactions between WCS and FA for the intakes of total and 18-carbon FA ($P \geq 0.29$); digestibility of DM, total FA, and 16-carbon and 18-carbon FA ($P \geq 0.23$); or the absorption of total FA and 18-carbon ($P \geq 0.17$; Table 2).

Overall, WCS increased the intake of DM by 1.95 kg/d ($P < 0.01$), NDF by 0.56 kg/d ($P < 0.01$), total FA by 443 g/d, 16-carbon FA by 123 g/d, and 18-carbon FA by 315 g/d ($P < 0.01$; Table 2). Although WCS did not affect 16-carbon FA digestibility ($P = 0.62$), it reduced total FA digestibility by 2.8 percentage units ($P = 0.01$) and 18-carbon FA digestibility by 4.9 percentage units ($P < 0.01$; Table 2). Additionally, WCS tended to reduce NDF digestibility by 2.8 percentage units ($P = 0.07$; Table 2). It also increased the absorption of total FA by 333 g/d, 16-carbon FA by 99.9 g/d, and 18-carbon FA by 230 g/d ($P < 0.01$; Table 2).

Overall, FA supplementation tended to reduce NDF intake by 0.44 kg/d ($P = 0.06$) and increased the intake of total FA by 352 g/d, 16-carbon FA by 284 g/d ($P < 0.01$), and 18-carbon FA intake by 58.1 g/d (all $P < 0.01$; Table 2). In addition, FA supplementation increased the absorption of total FA by 301 g/d ($P < 0.01$), 16-carbon FA by 235 g/d ($P < 0.01$), and 18-carbon FA by 60.0 g/d (all $P < 0.01$; Table 2).

Production Responses During the FR Period

Milk yield increased over time for all treatments (Figure 1B; Supplemental Table S2, see Notes), but no treatment differences were observed ($P = 0.19$; Table 3). Body weight and BCS decreased over time, but there was no effect of treatments ($P \geq 0.17$; Supplemental Figures S1A and S1B, see Notes). We did not observe any interactions between WCS and FA for the yields of milk protein and milk lactose, changes in BW and BCS, or feed efficiencies ($P \geq 0.41$; Table 3).

The interaction of WCS and FA supplementation affected the yields of milk fat ($P = 0.03$; Figure 2A), 3.5% FCM ($P = 0.09$; Supplemental Figure S1C), cumulative milk fat yield ($P = 0.10$), and milk fat content ($P = 0.02$; Table 3), and tended to affect the yield of ECM ($P = 0.13$; Figure 1C). Compared with CON, CT, FAT, and CT+FAT increased the yields of milk fat, 3.5% FCM, and ECM, as well as cumulative milk fat. No differences were observed between CT, FAT, and CT+FAT for these variables (Table 3). Additionally, CT+FAT increased milk fat content compared with the other treatments (Table 3).

Overall, WCS increased the yields of ECM by 3.72 kg/d ($P = 0.03$), 3.5% FCM by 4.13 kg/d ($P = 0.02$), milk fat by 0.15 kg/d ($P = 0.03$), and milk lactose by 0.17 kg/d ($P = 0.03$) and tended to increase milk protein by 0.08 kg/d ($P = 0.10$; Table 3). Overall, FA supplementation increased ECM/DMI by 0.15 kg/kg ($P = 0.04$), FCM/DMI by 0.18 kg/kg ($P = 0.03$), and milk fat content by

Table 2. Nutrient intake and total-tract nutrient digestibility for cows fed treatment diets during the FR period (d 1–21 postpartum)

Variable	Treatment ¹					P-value ²						
	CON	CT	FAT	CT+FAT	SEM	WCS	FA	WCS × FA	Time	WCS × time	FA × time	WCS × FA × time
Intake, kg/d												
DM	21.2 ^b	24.2 ^a	22.1 ^b	22.9 ^{ab}	0.77	<0.01	0.79	0.08	<0.01	0.54	0.45	0.31
NDF	6.73 ^b	7.65 ^a	6.65 ^b	6.86 ^b	0.24	<0.01	0.06	0.09	<0.01	0.47	0.38	0.40
Intake, g/d												
Total FA	413	823	733	1,206	28.2	<0.01	<0.01	0.29	—	—	—	—
16-carbon	85.2 ^d	179 ^c	340 ^b	492 ^a	10.9	<0.01	<0.01	<0.01	—	—	—	—
18-carbon	324	684	381	697	21.0	<0.01	<0.01	0.98	—	—	—	—
Digestibility, %												
DM	66.5	65.7	65.9	65.1	0.66	0.21	0.35	0.97	—	—	—	—
NDF	47.6	44.3	44.7	42.5	1.56	0.07	0.11	0.71	—	—	—	—
Total FA	85.1	81.1	84.2	82.7	1.01	0.01	0.73	0.23	—	—	—	—
16-carbon	81.3	80.3	82.1	82.0	0.99	0.62	0.22	0.64	—	—	—	—
18-carbon	87.7	81.8	87.3	83.4	1.15	<0.01	0.57	0.40	—	—	—	—
Absorbed FA, g/d												
Total FA	351	649	616	985	25.3	<0.01	<0.01	0.17	—	—	—	—
16-carbon	69.3 ^d	144 ^c	279 ^b	404 ^a	9.68	<0.01	<0.01	0.01	—	—	—	—
18-carbon	283	503	333	572	19.8	<0.01	<0.01	0.63	—	—	—	—

^{a–d}Means in a row with different superscripts differ. Shown only if a 2-way interaction between WCS and FA was observed ($P \leq 0.10$).

¹CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

²P-value refers to the ANOVA results for the main effect of WCS, the main effect of FA, the main effect of time, the interaction between WCS and FA, the interaction between WCS and time, the interaction between FA and time, and the interaction between WCS, FA, and time. A dash (—) indicates that the effects do not apply to those variables.

Table 3. Milk production, milk composition, BW, and BCS for cows fed treatment diets during the FR period (d 1–21 postpartum)

Variable	Treatment ¹					P-value ²						
	CON	CT	FAT	CT+FAT	SEM	WCS	FA	WCS × FA	Time	WCS × time	FA × time	WCS × FA × time
Yield, kg/d												
Milk	41.8	44.2	43.2	44.6	1.73	0.19	0.52	0.74	<0.01	0.34	0.40	0.55
Fat	1.85 ^b	2.15 ^a	2.12 ^a	2.12 ^a	0.07	0.03	0.07	0.03	<0.01	<0.01	0.90	0.97
Protein	1.47	1.59	1.50	1.54	0.06	0.10	0.87	0.43	<0.01	0.43	0.02	0.66
Lactose	2.01	2.24	2.11	2.22	0.08	0.03	0.51	0.41	<0.01	0.44	0.05	0.61
3.5% FCM ³	47.9 ^b	54.8 ^a	53.1 ^a	54.5 ^a	1.75	0.02	0.13	0.09	<0.01	0.01	0.58	0.91
ECM ⁴	48.2 ^b	54.3 ^a	52.3 ^a	53.6 ^a	1.72	0.03	0.27	0.13	<0.01	0.02	0.39	0.93
Content, %												
Fat	4.54 ^b	4.65 ^b	4.95 ^a	4.65 ^b	0.10	0.31	0.02	0.02	<0.01	<0.01	0.42	0.25
Protein	3.52	3.56	3.45	3.42	0.06	0.99	0.10	0.58	<0.01	0.37	0.73	0.34
Lactose	4.82	4.87	4.82	4.83	0.03	0.41	0.47	0.47	<0.01	0.66	0.94	0.97
FCM/DMI, kg/kg	2.27	2.19	2.50	2.33	0.08	0.13	0.03	0.62	0.08	0.29	0.89	0.52
ECM/DMI, kg/kg	2.27	2.18	2.46	2.31	0.08	0.11	0.04	0.64	0.05	0.27	0.86	0.41
BW, kg	718	750	719	729	17.0	0.17	0.51	0.45	<0.01	0.79	0.37	0.43
BW change, kg/d	−1.35	−1.42	−2.02	−1.63	0.35	0.64	0.20	0.49	—	—	—	—
BCS	3.16	3.16	3.08	3.20	0.04	0.19	0.62	0.18	<0.01	0.65	0.23	0.62
BCS change	−0.07	−0.06	−0.10	−0.08	0.02	0.46	0.15	0.72	—	—	—	—

^{a,b}Means in a row with different superscripts differ. Shown only if a 2-way interaction between WCS and FA was observed ($P \leq 0.10$).

¹CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

²P-value refers to the ANOVA results for the main effect of WCS, the main effect of FA, the main effect of time, the interaction between WCS and FA, the interaction between WCS and time, the interaction between FA and time, and the interaction between WCS, FA, and time. A dash (—) indicates that the effects do not apply to those variables.

³3.5% FCM = $(0.4324 \times \text{kg of milk}) + (16.216 \times \text{kg of milk fat})$; NRC, 2001).

⁴ECM = $(0.327 \times \text{kg of milk}) + (12.95 \times \text{kg of milk fat}) + (7.20 \times \text{kg of milk protein})$. This equation corrects milk to a 0.68 Mcal/kg energy basis.

0.20% ($P = 0.02$) and tended to increase milk fat yield by 0.12 kg/d ($P = 0.07$) and reduce milk protein content by 0.11% ($P = 0.10$; Table 3).

Milk FA Contents and Yields During the FR Period

Milk FA are derived from 2 sources: <16 carbon FA from de novo synthesis in the mammary gland and >16 carbon FA originating from extraction from plasma. Mixed-source FA (C16:0 and palmitoleic acid) originate from de novo synthesis in the mammary gland and extraction from plasma. No interactions were found between WCS and FA supplementation for the yield of milk FA sources ($P \geq 0.20$; Table 3) and most individual milk FA yields and contents ($P \geq 0.16$; Supplemental Tables S3 and S4, see Notes). On a yield basis, WCS increased the yield of preformed FA by 116 g/d ($P < 0.01$). Fatty acid supplementation increased the yield of mixed FA by 80 g/d ($P < 0.01$) and reduced the yield of de novo FA by 57 g/d ($P < 0.01$). The interaction of WCS and FA supplementation with time affected the content of mixed FA ($P = 0.08$; Table 4). Overall, WCS increased the content of preformed FA ($P = 0.02$) and reduced mixed FA ($P < 0.01$; Table 4). Fatty acid supplementation increased the content of mixed FA ($P < 0.01$) and reduced de novo FA ($P < 0.01$; Table 4).

Energy Intake and Energy Balance During the FR Period

During the FR period, WCS and FA supplementation tended to interact to affect milk energy output ($P = 0.15$). No differences in milk energy output were detected among CT, FAT, and CT+FAT; however, all 3 treatments tended to increase milk energy output compared with CON (Table 5). No interactions were found between WCS and FA for NEL intake, energy output for maintenance, body energy reserves, and energy balance ($P \geq 0.33$; Table 5). Overall, WCS increased NEL intake by 3.4 Mcal/d ($P < 0.01$). Overall, FA supplementation did not impact energy variables during the FR period ($P \geq 0.15$).

Plasma Insulin and Metabolites During the FR and CO Periods

During the FR period, WCS and FA supplementation interacted with time to affect plasma NEFA ($P = 0.10$; Table 6). In the second week of lactation, FAT increased plasma NEFA, and CT tended to increase plasma NEFA compared with CON ($P = 0.02$ and $P = 0.10$, respectively; Supplemental Table S2). The interaction of WCS and FA supplementation affected plasma BHB ($P < 0.08$). Spe-

Table 4. Summation of milk FA content and yield for cows fed treatment diets during the FR period (d 1–21 postpartum)

Variable ¹	Treatment ²					P-value ³						
	CON	CT	FAT	CT+FAT	SEM	WCS	FA	WCS × FA	Time	WCS × time	FA × time	WCS × FA × time
Summation by source, g/100 g of FA												
De novo	22.7	21.2	18.8	17.6	0.96	0.13	<0.01	0.86	<0.01	0.07	0.38	0.57
Mixed	31.6	30.4	34.1	32.7	0.39	<0.01	<0.01	0.73	0.74	0.42	1.00	0.08
Preformed	45.7	48.4	47.0	49.7	1.23	0.02	0.30	0.97	<0.01	<0.05	0.55	0.62
Summation by source, g/d												
De novo	406	422	365	349	18.1	0.99	<0.01	0.32	<0.01	0.71	0.35	0.86
Mixed	556	603	664	656	22.9	0.36	<0.01	0.20	<0.01	0.64	0.78	0.65
Preformed	799	962	920	988	49.4	<0.01	0.11	0.26	<0.01	0.10	0.93	0.73

¹De novo FA originate from mammary de novo synthesis (<16 carbons), preformed FA originate from extraction from plasma (>16 carbons), and mixed FA originate from both sources (C16:0 plus C16:1 *cis*-9). Concentrations and yields of individual FA are reported in Supplemental Tables S3 and S4, respectively.

²CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

³P-value refers to the ANOVA results for the main effect of WCS, the main effect of FA, the main effect of time, the interaction between WCS and FA, the interaction between WCS and time, the interaction between FA and time, and the interaction between WCS, FA, and time.

cifically, CT and FAT tended to increase plasma BHB compared with CON, but their effect did not differ from that of CT+FAT (Table 6). Whole cottonseed and time tended to interact to modify plasma insulin ($P = 0.13$; Table 6). In the second week of lactation, WCS diets decreased plasma insulin compared with non-WCS diets ($P = 0.02$; Supplemental Table S2). Overall, WCS and FA increased plasma gossypol and (–) and (+) isomers (P

≤ 0.02 ; Table 7). During the CO period, WCS increased plasma gossypol and (–) and (+) isomers ($P < 0.01$; Table 7).

Production Responses During the CO Period

Whole cottonseed and FA supplementation interacted with time to affect the yields of ECM ($P = 0.03$; Figure

Table 5. Energy intake, energy output and balance, and efficiency of energy utilization for cows fed treatment diets during the FR period (d 1–21 postpartum)

Variable	Treatment ¹					P-value ²		
	CON	CT	FAT	CT+FAT	SEM	WCS	FA	WCS × FA
NEL intake, Mcal/d ³	30.2	34.7	31.0	33.3	1.10	<0.01	0.77	0.30
Milk ⁴	34.7	37.9	37.7	37.6	1.34	0.19	0.24	0.15
Maintenance ⁵	13.8	14.4	13.8	14.0	0.26	0.11	0.33	0.42
Body reserves ⁶	–7.91	–7.25	–12.8	–8.20	2.22	0.19	0.15	0.33
Energy balance ⁷	–18.5	–17.6	–20.5	–18.3	1.27	0.19	0.27	0.58

¹CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

²P-value refers to the ANOVA results for the main effect of WCS, the main effect of FA, and the interaction between WCS and FA.

³Calculated with the NASEM (2021) equations for predicting dietary digestible energy, ME, and NEL (Mcal/kg DM). We incorporated all dietary composition data (Table 1), DMI, and digestibilities of NDF and FA (Table 2); all other parameters were estimated through the NASEM (2021) software (V8 R2024.05.05).

⁴Milk NEL (Mcal/d) = milk yield (kg/d) × [(fat % × 0.0929) + (true protein % × 0.0585) + (lactose % × 0.0395)] (NASEM, 2021).

⁵NEL maintenance (Mcal/d) = 0.10 Mcal/kg × BW (kg)^{0.75} (NASEM, 2021).

⁶Body reserves NEL (Mcal/d) = 5.6 × BW change (NASEM, 2021).

⁷Energy balance (Mcal/d) = NEL intake (Mcal/d) – milk NEL (Mcal/d) – NEL maintenance requirement (Mcal/d).

Table 6. Blood variables for cows fed treatment diets during the FR period (d 1–21 postpartum)

Variable ¹	Treatment ¹					P-value ²						
	CON	CT	FAT	CT+FAT	SEM	WCS	FA	WCS × FA	Time	WCS × time	FA × time	WCS × FA × time
Insulin, μ IU	4.43	3.89	3.96	3.61	0.28	0.12	0.20	0.73	<0.01	0.13	0.93	0.54
NEFA, mmol/L	0.60	0.72	0.71	0.75	0.07	0.23	0.33	0.55	<0.01	0.29	0.12	0.10
BHB, mg/dL	8.65	10.5	10.6	9.70	0.79	0.56	0.45	0.08	0.76	0.06	0.27	0.77

¹CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

²P-value refers to the ANOVA results for the main effect of WCS, the main effect of FA, the main effect of time, the interaction between WCS and FA, the interaction between WCS and time, the interaction between FA and time, and the interaction between WCS, FA, and time.

1C), 3.5% FCM ($P = 0.06$; Supplemental Figure S1C), milk protein ($P = 0.01$; Figure 2B), and milk lactose ($P < 0.01$; Figure 2C) and tended to alter milk fat content ($P = 0.14$; Table 8), but when we compared treatments within a week of lactation, we did not observe differences ($P > 0.10$; Supplemental Table S5, see Notes). Additionally, we found a 3-way interaction between WCS, FA, and time for ECM/DMI and a tendency for FCM/DMI ($P = 0.06$ and $P = 0.12$). In particular, FAT tended to increase ECM/DMI and FCM/DMI compared with CON in wk 4, and FAT tended to increase ECM/DMI compared with CT in wk 4. Supplemental FA and time interacted to influence BW ($P = 0.07$; Table 8), but when we compared the effect of supplemental FA within a week of lactation, we did not observe differences ($P > 0.17$; Supplemental Table S5). Overall, WCS tended to increase cumulative milk protein yield by 1.7 kg ($P = 0.07$; Table 8). Overall, FA supplementation tended to increase cumulative milk protein yield by 1.7 kg ($P = 0.08$) and tended to reduce BW loss by 0.46 kg/d ($P = 0.06$) and milk fat content by 0.21 percentage units ($P = 0.06$; Table 8).

DISCUSSION

The inclusion of FA supplements and oilseeds in dairy cow diets is a well-established strategy to increase energy intake (Rabiee et al., 2012; Lock et al., 2025). The inclusion of WCS in postpeak dairy cow diets has been shown to influence nutrient digestibility and milk production responses (Bales et al., 2024a; Adeniji et al., 2025). However, prior research evaluating WCS in early-lactation cows, either alone or in combination with FA supplementation, has only begun from the third week postpartum (Harrison et al., 1995; Abel-Caines et al., 1997; Chen et al., 2008). To our knowledge, no studies have examined the effect of WCS inclusion immediately postcalving. Regarding supplemental FA, feeding C16:0 and C18:1 *cis*-9 FA blends during this period can positively affect milk production responses (de Souza et al., 2021). Given the potential for WCS and FA supplementation to influence production outcomes, we aimed to evaluate their individual and combined effects in the immediate postpartum period—a time of high metabolic demand and rapid physiological change. Therefore, we evaluated the effect of feeding supplemental FA, using either WCS or a C16:0 and C18:1 *cis*-9 FA supplement, on nutrient

Table 7. Plasma gossypol¹ for cows fed treatment diets during the FR (d 1 to 21 postpartum) and CO (d 22 to 42 postpartum) periods

Item	Treatment ²				P-value ³		
	CON	WCT	FAT	WCT+FAT	WCS	FA	WCS × FA
Fresh period							
Total gossypol	0.36 (0.34–0.38)	1.14 (1.08–1.19)	0.38 (0.37–0.40)	1.22 (1.15–1.28)	<0.01	0.01	0.91
(-) gossypol	0.20 (0.18–0.21)	0.82 (0.76–0.89)	0.22 (0.21–0.24)	0.88 (0.81–0.95)	<0.01	0.01	0.34
(+) gossypol	0.32 (0.31–0.34)	0.86 (0.83–0.90)	0.33 (0.32–0.34)	0.92 (0.88–0.97)	<0.01	0.03	0.27
Carryover period							
Total gossypol	0.32 (0.30–0.34)	0.65 (0.60–0.70)	0.33 (0.30–0.35)	0.65 (0.61–0.70)	<0.01	0.83	0.82
(-) gossypol	0.20 (0.18–0.22)	0.50 (0.45–0.55)	0.19 (0.18–0.21)	0.50 (0.45–0.55)	<0.01	0.84	0.99
(+) gossypol	0.27 (0.25–0.29)	0.46 (0.43–0.49)	0.28 (0.26–0.29)	0.47 (0.44–0.50)	<0.01	0.55	0.65

¹Plasma gossypol data lacked a normal distribution and were transformed using a $\log_{10}$ function. Back-transformed values are reported as LSM (95% CI).

²CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

³P-value refers to the ANOVA results for the main effect of WCS, the main effect of FA, and the interaction between WCS and FA.

Table 8. Dry matter intake, milk production, milk composition, BW, and BCS for cows fed treatment diets during the CO period (d 22–42 postpartum)

Item	Treatment ¹					P-value ²						
	CON	CT	FAT	CT+FAT	SEM	WCS	FA	WCS × FA	Time	WCS × time	FA × time	WCS × FA × time
DMI	25.9	27.0	26.2	26.2	0.80	0.39	0.73	0.38	<0.01	0.68	0.38	0.81
Yield, kg/d												
Milk	53.0	53.8	54.7	55.5	2.36	0.63	0.31	0.98	<0.01	0.22	0.76	0.23
Fat	1.96	2.03	1.97	2.00	0.08	0.52	0.86	0.81	0.03	0.26	0.32	0.19
Protein	1.58	1.64	1.69	1.69	0.06	0.18	0.20	0.90	<0.01	0.54	0.19	0.01
Lactose	2.61	2.67	2.69	2.76	0.11	0.44	0.28	0.99	<0.01	0.12	0.52	<0.01
3.5% FCM ³	54.9	56.3	55.3	56.7	2.18	0.40	0.81	0.99	0.03	0.19	0.41	0.06
ECM ⁴	54.2	55.8	54.9	56.4	2.07	0.34	0.67	0.99	0.01	0.20	0.45	0.03
Content, %												
Fat	3.74	3.78	3.60	3.50	0.11	0.77	0.06	0.53	<0.01	0.73	0.03	0.14
Protein	3.02	3.05	3.01	3.02	0.06	0.77	0.78	0.80	0.35	0.63	0.08	0.96
Lactose	4.90	4.94	4.92	4.90	0.03	0.73	0.62	0.36	0.35	0.23	0.61	0.58
FCM/DMI, kg/kg	2.10	2.09	2.11	2.08	0.06	0.74	0.95	0.83	<0.01	0.36	0.08	0.12
ECM/DMI, kg/kg	2.07	2.07	2.10	2.09	0.06	0.92	0.58	0.97	<0.01	0.21	0.08	0.06
BW, kg	709	732	701	711	16.1	0.23	0.33	0.65	<0.01	0.28	0.02	0.78
BW change, kg	−0.69	−0.62	−0.28	−0.10	0.24	0.60	0.06	0.81	—	—	—	—
BCS	3.00	2.97	2.87	2.93	0.06	0.82	0.20	0.48	<0.01	0.76	0.56	0.91
BCS change	−0.03	−0.03	−0.04	−0.02	0.02	0.70	0.84	0.81	—	—	—	—

¹CON = a diet containing no WCS or supplemental FA; CT = diet containing WCS at 10% diet DM; FAT = diet supplemented with a calcium salt containing 60% C16:0 and 30% C18:1 *cis*-9; CT+FAT = diet containing WCS at 10% diet DM and a C16:0 C18:1 *cis*-9 calcium salt.

²P-value refers to the ANOVA results for the main effect of WCS, the main effect of FA, the main effect of time, the interaction between WCS and FA, the interaction between WCS and time, the interaction between FA and time, and the interaction between WCS, FA, and time. A dash (—) indicates that the effects do not apply to those variables.

³3.5% FCM = (0.4324 × kg of milk) + (16.216 × kg of milk fat; NRC, 2001).

⁴ECM = (0.327 × kg of milk) + (12.95 × kg of milk fat) + (7.20 × kg of milk protein). This equation corrects milk to a 0.68 Mcal/kg energy basis.

digestibility and milk production in early-lactation cows. Our results provide evidence that feeding either WCS or supplemental FA improved milk production responses. However, feeding WCS and supplemental FA together did not yield additive benefits and instead moderated the positive responses observed when fed separately.

It has been suggested that feeding supplemental FA in the immediate postpartum period may depress feed intake (Kuhla et al., 2016), potentially due to hepatic oxidation of fuels regulating DMI (Allen and Piantoni, 2013). Grummer (1992) reported that fat supplements with a high content of unsaturated FA (e.g., tallow or heat-treated soybeans) had limited benefit to milk production responses in early-lactation cows, likely due to the total dietary FA content (5%–6% DM), the unsaturation level of the FA supplements used, and timing of supplementation. In contrast, feeding C16:0 and C18:1 *cis*-9 blends at 1.5% DM improved milk production responses without compromising DMI or body energy reserves (de Souza et al., 2021; Lock et al., 2025). The absence of an additive effect when WCS and FA were fed together suggests that their individual benefits on the yields of milk fat and ECM may not be complementary, potentially due to limitations in intake or digestibility.

The lack of effect on milk production responses from feeding WCS and supplemental FA together compared with feeding them individually in our study partially

aligns with previous findings (Harrison et al., 1995). One likely explanation is the reduced NDF digestibility observed in the CT+FAT treatment; this reduction could be due to multiple factors, including the lower fiber digestibility of WCS compared with soyhulls (NASEM, 2021) and the inhibitory effects of UFA on rumen cellulolytic bacteria (Maia et al., 2007, 2010; Hanno et al., 2024; Sears et al., 2024). Reduced NDF digestibility can limit acetate availability, a key precursor for *de novo* FA synthesis in the mammary gland (Urrutia et al., 2019). dos Santos Neto et al. (2023) similarly reported reduced NDF digestibility and no improvement in milk fat yield when infusing C18:1 *cis*-9 with emulsifiers. Thus, both the lower fiber quality of WCS and the inhibitory effects of UFA likely contributed to the lack of additive production responses when WCS and FA were fed together.

Regarding FA digestibility, the primary FA in WCS, linoleic acid, is mostly biohydrogenated to stearic acid (C18:0) in the rumen (Palmquist et al., 2005), increasing C18:0 duodenal flow and reducing FA digestibility (Boerman et al., 2015, 2017; Burch et al., 2021). Moreover, total FA digestibility decreases as duodenal FA flow increases (Boerman et al., 2015). We observed greater 18-carbon FA intake in CT and CT+FAT, leading to increased C18:0 duodenal flow and reduced 18-carbon and total FA digestibility, consistent with previous studies in postpeak cows (de Souza et al., 2018; Pierce et al., 2024;

Bales et al., 2024a). Although WCS increased DMI, this response was moderated when WCS was fed alongside supplemental FA, likely due to increased duodenal flow of FA stimulating gut peptides that suppress intake (Bradford et al., 2008). The greater intake of 18-carbon UFA, differences in NDF structure between WCS and soyhulls, reductions in NDF and FA digestibility, and effect on DMI likely explain the lack of effect on milk production responses when feeding WCS in combination with supplemental FA. Further research is required to determine if the combined use of WCS and FA supplementation can improve animal performance.

Milk production responses in dairy cows are affected by nutrient flow to the mammary gland (Davis and Collier, 1985). Increasing absorbed FA may enhance mammary uptake of preformed FA while reducing de novo FA synthesis (Glasser et al., 2008; dos Santos Neto et al., 2021b). The increases in 3.5% FCM and ECM observed in the current study were mainly driven by higher milk fat yield, resulting from increased yields of preformed and mixed FA when WCS and FA were fed. These findings are consistent with previous studies in mid-lactation cows in which WCS was replaced with soyhulls (Rico et al., 2017; Bales et al., 2024a), in cows fed WCS from wk 3 to 15 postpartum (Harrison et al., 1995), and in early-lactation cows supplemented with C16:0-enriched or C16:0 and C18:1 *cis*-9 FA blends (de Souza and Lock, 2019; de Souza et al., 2021; Parales-Girón et al., 2025). Feeding WCS likely increased milk fat and preformed FA yield through higher DMI and greater 18-carbon FA absorption (Harrison et al., 1995; Rabiee et al., 2012), while supplemental FA increased milk fat and mixed FA yields primarily due to greater C16:0 absorption. In the mammary gland, C16:0 plays a key role in milk triglyceride formation by enhancing glycerol-3-phosphate acyltransferase activity (Tzompa-Sosa et al., 2014), which has a higher preference for C16:0 as a substrate to initiate milk triglyceride synthesis compared with other FA (Kinsella and Gross, 1973).

Recent data demonstrate the efficacy of supplemental C16:0 and C18:1 *cis*-9 in the initial 3 wk postpartum. Feeding a C16:0-enriched supplement increased the yields of 3.5% FCM and ECM but tended to increase BW loss compared with a nonsupplemental control diet (de Souza and Lock, 2019). This outcome partially agrees with a recent study feeding a C16:0-enriched supplement and chromium in which C16:0 increased the yields of 3.5% FCM and ECM without affecting BW loss (Parales-Girón et al., 2025). A previous study reported positive effects of supplementing C16:0 and C18:1 *cis*-9 on the yields of milk fat and ECM compared with a nonsupplemental FA control diet, with increasing C18:1 *cis*-9 content in the supplemental FA reducing BW loss (de Souza et al., 2021). In the current study, although we did not observe a

main effect of FA supplementation on the yields of 3.5% FCM and ECM, we observed improvements in the yields of 3.5% FCM and ECM with FA supplementation, with CT, FAT, and CT+FAT increasing ECM yield by 6.1, 4.1, and 5.4 kg/d, respectively, compared with CON. The significant response observed for the CT treatment on milk production responses could have masked the main effect of FA supplementation.

Regarding BW loss, although we observed an increased ECM yield for CT, FAT, and CT+FAT compared with CON, these treatments did not affect BW and BCS loss during the FR period. These results differed from our previous work in which we observed increased ECM yield with a linear reduction in BW loss reduction with increasing dietary C18:1 *cis*-9 (de Souza et al., 2021) and a tendency for increased body energy reserves mobilization with increased ECM yield associated with supplementation of C16:0 in early-lactation cows (de Souza and Lock, 2019). Cows with higher BCS at parturition are more susceptible to greater BW and BCS losses, increased risk of metabolic disorders, and reduced production and reproductive performance in the immediate postpartum (Roche et al., 2009). In our previous studies evaluating the effects of increasing dietary C18:1 *cis*-9 content and C16:0 supplementation, the average precalving BCS was ~3.75 (de Souza and Lock, 2019; de Souza et al., 2021), whereas in the current study, it was 3.36. The lower average BCS in the current study may explain the differences observed in BW and BCS loss compared with our previous reports.

Gossypol is a polyphenolic compound found in the pigment glands of WCS that has been reported to reduce fertility and negatively affect embryo development in dairy cows (Santos et al., 2003). Although our study was not designed to evaluate possible effects on reproductive performance, we analyzed plasma for (+) and (–) isomers and total gossypol. We observed an effect of WCS increasing plasma gossypol, which agrees with previous studies feeding WCS to early-lactation and postpeak cows (Santos et al., 2002; Bales et al., 2024a). The observed effect of FA supplementation was due to the WCS in the CT+FAT treatment, because plasma gossypol levels were similar for CT compared with CT+FAT, although CON and FAT had the lowest values and were not different. It is important to note that the values observed in our study were below the 5.0 µg/mL of plasma gossypol threshold indicated as safe (Mena et al., 2001). Similar to our study, Santos et al. (2002) reported that elevated plasma gossypol levels up to 5.9 µg/mL did not affect milk production responses. Additionally, we observed a carryover effect of WCS increasing plasma gossypol, with lower values than those observed during the FR period. This finding agrees with the results reported by Mena et al. (2001), in which 2 wk of feeding a gossypol-free diet lowered

plasma gossypol, but levels were still higher than in cows that were not fed WCS.

One of the objectives of our study was to determine possible carryover effects of the treatment diets. Nutritional interventions in the immediate postpartum period have been reported to influence animal performance even after they have ceased (Jørgensen et al., 2016; de Souza et al., 2021; Parales-Girón et al., 2025). Although no persistent effects on overall ECM yield were observed, milk protein yield was increased by CT+FAT compared with CON during certain weeks of the CO period, and both WCS and FA supplementation increased cumulative protein yield by 1.7 kg. The FA supplementation also reduced BW loss during the CO period, suggesting a shift in energy partitioning toward body reserves. Supplemental C18:1 *cis*-9 may enhance mitochondrial function and insulin sensitivity in adipose tissue by increasing PLIN5 and PPAR α protein expression and mitochondrial density (Abou-Rjeileh et al., 2023). The PLIN5 facilitates lipid droplet-mitochondria interactions to support lipid mobilization (Zhang et al., 2022), while PPAR α regulates genes involved in FA oxidation and mitochondrial biogenesis (Bougarne et al., 2018). These molecular changes suggest a shift in adipose tissue metabolism that may favor nutrient storage and recovery of body condition. As lactation progresses and milk output stabilizes, such adaptations could reflect a broader homeorhetic response redirecting nutrient partitioning away from milk synthesis and toward replenishment of maternal reserves, particularly as negative energy balance diminishes. The absence of a milk yield carryover effect, in contrast with findings from de Souza et al. (2021), may reflect differences in body reserve status and diet composition. Notably, our CO period diet excluded WCS and had a lower FA content (1.93% DM vs. 2.94% DM), which may have limited the persistence of treatment effects. Some studies have reported no carryover effects on milk yield despite early postpartum nutritional interventions (e.g., Lee et al., 2019; Albornoz et al., 2024), suggesting that the magnitude and duration of such effects may depend on the interaction between diet, physiological state, and metabolic flexibility (Nayeri and Stothard 2016). These findings highlight the potential for early postpartum nutrition to influence longer-term energy balance, although the precise mechanisms underlying FA carryover effects on BW change and milk production remain unclear and warrant further investigation.

Our study aim was to determine the effects of feeding WCS and FA to early-lactation cows. The treatment diets were formulated to have similar CP, starch, NDF, and forage NDF. However, WCS and supplemental FA increased dietary FA, resulting in nonisoenergetic treatment diets. Whereas we could have chosen to include a

different FA supplement (e.g., C18:0-enriched FA prills), increasing the FA content in CON using an FA supplement could influence animal performance because of the different metabolic effects of long-chain FA (de Souza et al., 2018; dos Santos Neto et al., 2021a,b; Bales et al., 2024b). Additionally, our previous research has shown that increasing dietary energy using a C18:0-enriched FA supplement may not increase energy intake (Western et al., 2020) and may negatively affect milk production responses (Piantoni et al., 2015). Given our focus on determining the effect of increasing dietary FA from either WCS or a C16:0 and C18:1 *cis*-9 supplement, we did not include an FA supplement in the CON treatment and did not formulate the diets to be isoenergetic. One final consideration is that we replaced soyhulls and soybean meal to balance NDF and CP across diets when adding WCS in our study. We acknowledge that NDF and protein in WCS differ from those in soybean by-products, which may have affected our results. In previous studies, we found similar increases in milk production responses when we replaced cottonseed hulls and meal with WCS (Bales et al., 2024b) or soybean by-products with WCS (Rico et al., 2017; de Souza et al., 2018). These increases were observed regardless of the by-products used to balance the diets' nutrient specifications. Although replacing cottonseed hulls and meal with WCS would be a cleaner approach from a research perspective, the substitutions in our treatment diets are more relevant to the industry and can be implemented immediately on dairy farms because soybean by-products are readily available and commonly used in dairy rations.

CONCLUSIONS

To our knowledge, no other studies have evaluated the effect of both WCS and a 60% C16:0 and 30% C18:1 *cis*-9 FA supplement in the immediate postpartum of dairy cows. In this study, CT, FAT, and CT+FAT increased milk fat and 3.5% FCM yields and tended to increase ECM by 0.30, 0.27, and 0.27 kg/d; 6.9, 5.2, and 6.6 kg/d; and 6.1, 4.1, and 5.4 kg/d, respectively, compared with the control diet. Additionally, treatment diets did not compromise body energy reserves or BW loss during this period. The absence of a synergistic or additive effect when feeding both WCS and supplemental FA could be explained by the lack of effect on DMI and reduced NDF and FA digestibility. These results indicate that both WCS and supplemental FA in the diet can improve milk production of early-lactation dairy cows, with WCS improving milk fat and protein yields and supplemental FA improving milk fat yield.

NOTES

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Nonstandard abbreviations used: C16:0 = palmitic acid; C18:0 = stearic acid; C18:1 *cis*-9 = oleic acid; CO = carryover; CON = diet containing no WCS or supplemental FA; CT = diet containing WCS; CT+FAT = diet containing WCS and the C16:0 *cis*-9 C18:1 calcium salt; FA = fatty acids; FAT = diet supplemented with a calcium salt containing (% of total FA) 60% C16:0 and 30% C18:1 *cis*-9; FR = fresh; NEFA = nonesterified fatty acids; WCS = whole cottonseed.

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